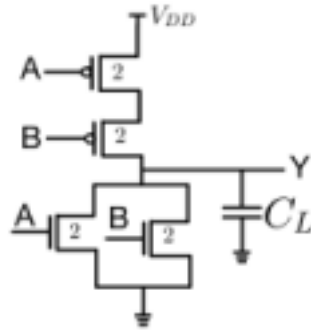


BRAC University
Department of Computer Science and Engineering
CSE 460, Quiz 5, Fall 2025

Name:

Student ID:

Q1.



A digital system-on-chip has the above gate from the figure implemented on it. Individual transistor widths $k_{nMOS} = 2$, $k_{pMOS} = 2$ were chosen to achieve the effective rise and fall resistance equal to that of a unit inverter, R for the given gate. The system has $V_{DD} = 1V$, $\lambda = 150$ nm, load capacitor, $C_L = 2pF$, gate and parasitic diffusion capacitance for both PMOS and NMOS transistors are 0.4 fF/ μm and 0.5 fF/ μm respectively. The node Y switches once in every 1 ns.

- (i) Calculate the total capacitance at node Y.
- (ii) Estimate the Switching power when operating frequency is 2 GHz.
- (iii) Determine if the system can switch faster or slower, when the switching power is halved while V_{DD} and the capacitance remain same [as (ii)].